

XU BOTIAN

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OBJECTIVE

To become skilled and proficient in the engineering field. I aim to apply my skills to this diverse workplace, demonstrating a strong focus on quality and a passion for success.

EDUCATION

Apr 2023 – May 2026

Diploma in Mechatronics with Merit

Temasek Polytechnic

- Achieved cGPA of 3.96.
- Achieved Course Silver Medal.
- Achieved Temasek Poly Director's List Award twice.
- Completed modules in Statics and Strength of Materials, Electronic Prototyping, Electronic Device and Circuits, Engineering Drawing, Control and Automation among others.

Apr 2021 – Mar 2023

Higher NITEC in Mechanical Engineering

Institute of Technical Education

- Completed modules in Engineering Materials and Mechanics, Quality Engineering, Cobot Automation in Manufacturing among others.

SKILLS

Technical/Domain Skills

- Proficient in all forms of Microsoft Office service, i.e. PowerPoint, Word, Excel among others.
- Proficient in 3D Modelling software, such as SolidWorks and CREO.
- Experience in Mechanical Design.
- Experience in basic Robotic system programming.
- Experience in design and patch circuit in pneumatic systems.

Generic Skills

- Strong communication in both English and Mandarin.
- Effective in collaborating with colleagues well.
- Work well under pressure.
- Logical thinker with a focus on efficient problem-solving.
- Maintain a positive attitude and calm demeanour.
- Strong Hands-on skills.
- Keen to join company events.

WORK EXPERIENCE

July 2023 – Apr 2026

Part Time Service Crew

Makino Jewel, EN Group, En Holdings Pte. Ltd.

- Perform strong communication skills while being cashier and hosting.

Apr 2025 – Sep 2025

Internship

Healthcare Engineering Centre (HEC), Temasek Polytechnic

- Applied strong hands-on skills to build and integrate a Smart Agrivoltaics Urban Farming system, setting up LED lighting, misting water circulation, and solar panels.
- Demonstrated creative problem-solving and technical adaptability by redesigning electrical circuits with heat sinks to resolve critical driver overheating issues.
- Fabricated optics, microfluidic modules, and enclosures for a colorimetric sensing prototype using 3D printing (CAD) and laser cutting technologies.
- Gained practical knowledge in IoT by configuring Bluetooth communication protocols for wireless control dashboards and conducting precision spectral testing.
- Performed strong communication and presentation skills by co-managing a project booth at

the NTU 'Live it Up' event at Marina Bay to demonstrate prototypes to industry professionals.

Oct 2022 – Feb 2023

Internship

Urban Green Dot Pte. Ltd.

- Applied strong hands-on skills to build aluminium profiles, the foundational structure of hydroponic systems.
- Demonstrated patience, persistence, and hand-eye coordination by cutting, gluing, and connecting channels and tubes for hydroponic systems.
- Gained knowledge of materials, farming, and hydroponic systems by participating in the entire hydroponic farming process, including plant fertilization, measuring EC levels, and more.
- Perform collaboration and communication skills when taking part in workshops and set up a booth at The Singapore Institute of Management (SIM) to promote company and products.

AWARDS/ACHIEVEMENTS

2026

- Excel Marco Industrial Systems Course Silver Medal (Second best student of the course).

2025

- Temasek Poly Director's List Award for the academic year 2024/25 (Top 10% of the cohort).

2024

- Temasek Poly Director's List Award for the academic year 2023/24 (Top 10% of the cohort).

2022

- Complete WHO Standard Operating Procedures (SOPs) for Emergencies.
- Completion of SMART INDUSTRY READINESS INDEX (SIRI) AWARENESS COURSE (2h).

2020

- Silver Level of National Youth Achievement Award (NYAA).
- Young Engineer Award (Bronze).

2019

- Academic Excellence Award in term 1.
- Academic Excellence Award in semester 1.
- Academic Excellence Award of Additional Mathematics (NA).
- Small Business Operations (30 Hours).
- Complete Outward Bound Course Singapore.

2018

- Gold award of Singapore Math Kangaroo Contest 2018.

CO-CURRICULAR ACTIVITIES

Apr 2024 – Apr 2025

Head of Logistic

Society of Mechatronics Engineers (SME club)

- Applied good budgeting and financial planning skills when I am being head of logistic and in charge of majority of the logistic stuff (pre-event documents, item lists and so on) before and after MTN Bonding Day.
- Applied strong organizational and time management skills as a coordinator of the MTN Bonding Day, a BBQ night event at Pasir Ris Park with over 50 participants.
- Passionate to learn new skills and applied 3D modelling skills while took part in Lazer cutting workshops.

REFERENCES

Available upon request.